

RETAIL ANALYTICS DECISION SUPPORT SYSTEM

Use Case Specification

1. UC-01. Analyze Sales Performance

Primary Actor: Sales Manager

Goal: Evaluate sales performance and compare revenue trends across different years to support sales planning and performance assessment.

Preconditions:

- The Sales Manager has access to the RADSS.
- Historical sales data is available in the system.

Main Flow:

1. The Sales Manager opens the Sales Analytics module.
2. The system displays the current sales performance overview.
3. The Sales Manager selects the year or years to be analyzed.
4. The system calculates and displays the corresponding revenue trends.
5. The Sales Manager compares sales performance across the selected years.
6. The Sales Manager uses the analysis to assess sales performance and support planning decisions.

Alternative Flow: If sales data is unavailable for the selected period, the system displays a message indicating that no data is available.

Postconditions: The Sales Manager has evaluated sales performance and identified relevant revenue trends for planning and management purposes.

2. UC-02. Generate Seasonal Sales Forecast

Primary Actor: Sales Manager

Goal: Generate a sales forecast for upcoming periods based on historical seasonal patterns to support sales planning.

Preconditions:

- The Sales Manager has access to the RADSS.
- Sufficient historical sales data is available for forecasting.

Main Flow:

1. The Sales Manager opens the Sales Analytics module.
2. The Sales Manager selects the forecasting period.
3. The system analyzes historical sales patterns and recurring seasonal fluctuations.
4. The system generates the sales forecast for the selected future period.

5. The system displays the forecast together with relevant historical trends.
6. The Sales Manager reviews the forecast and uses it to support upcoming sales planning.

Alternative Flow: If insufficient historical data is available, the system informs the Sales Manager that a reliable forecast cannot be generated.

Postconditions: The Sales Manager has a forecast of expected sales for the upcoming period and can use it as a basis for sales planning.

3. UC-03. Analyze Customer Value

Primary Actor: Marketing Manager

Goal: Analyze customer value and purchasing behavior to identify high-value customer segments and support customer management decisions.

Preconditions:

- The Marketing Manager has access to the RADSS.
- Customer transaction data is available in the system.

Main Flow:

1. The Marketing Manager opens the Customer Analytics module.
2. The system calculates customer value indicators based on revenue, purchase frequency, and average order value.
3. The system ranks customers according to their revenue contribution.
4. The Marketing Manager selects the customer segmentation criteria.
5. The system segments customers based on revenue and purchase frequency.
6. The system identifies high-value customers, including customers with high revenue but relatively low purchase frequency.
7. The Marketing Manager reviews the customer segments and value profiles to support customer management decisions.

Alternative Flow: If insufficient customer transaction data is available, the system informs the Marketing Manager that customer value analysis cannot be completed.

Postconditions: The Marketing Manager has identified customer value segments and relevant high-value customer profiles for customer management and marketing planning.

4. UC-04. Evaluate Product Performance

Primary Actor: Product Manager

Goal: Evaluate product performance across revenue and sales volume to identify core products and support product portfolio decisions.

Preconditions:

- The Product Manager has access to the RADSS.
- Product transaction data is available in the system.

Main Flow:

1. The Product Manager opens the Product Analytics module.
2. The system calculates product performance based on revenue and quantity sold.
3. The system ranks products by revenue contribution and sales volume.
4. The Product Manager reviews the product performance analysis.
5. The system identifies products that perform strongly across both revenue and sales volume.
6. The Product Manager reviews the identified core products to support product prioritization decisions.

Alternative Flow: If insufficient product transaction data is available, the system informs the Product Manager that product performance cannot be reliably evaluated.

Postconditions: The Product Manager has evaluated product performance and identified core products for product portfolio planning.

5. UC-05. Identify Promotion Candidates

Primary Actor: Product Manager

Goal: Identify products suitable for promotional consideration based on their performance indicators.

Preconditions:

- The Product Manager has access to the RADSS.
- Product performance data is available in the system.

Main Flow:

1. The Product Manager opens the Product Analytics module.
2. The Product Manager requests promotion candidate analysis.
3. The system evaluates product performance indicators.
4. The system identifies products that meet the criteria for promotional consideration.
5. The system displays the identified promotion candidates.
6. The Product Manager reviews the candidates to support promotional planning.

Alternative Flow: If no products meet the criteria, the system informs the Product Manager that no promotion candidates have been identified.

Postconditions: The Product Manager has a list of products suitable for promotional consideration.

